LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - III

Calculus - Mean Value Theorems

Topics covered:

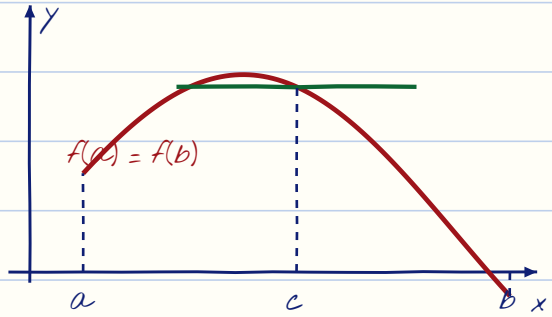
- Rolle's theorem and geometrical interpretation
- Lagrange's mean value theorem and geometrical interpretation
- Cauchy's mean value theorem
- Taylor's and Maclaurin's theorems with remainders (without proof)
- Problems and applications on the above theorems

Handwritten Notes - B.Tech (JNTUA R23)

1. Rolle's Theorem

Statement: if $f(x)$ is (i) continuous on the closed interval $[a, b]$, (ii) differentiable on the open interval (a, b) and (iii) $f(a) = f(b)$, then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Geometrical meaning: if the ordinates at the end points are equal, the graph has at least one point where the tangent is parallel to the x-axis (horizontal).



Rolle: $f(a) = f(b)$

there is a point c where the tangent is PARALLEL to the x-axis : $f'(c) = 0$

Method to verify: (1) check the three conditions ; (2) solve $f'(c) = 0$; (3) show c lies in (a, b).

Problem 1: verify Rolle's theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

Polynomial – continuous and differentiable. $f(1) = 0 = f(3)$. $f'(x) = 2x - 4 = 0 \Rightarrow c = 2 \in (1, 3)$ ✓.

Problem 2: verify Rolle's theorem for $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$.

Continuous and differentiable everywhere. $f(\pi/4) = e^{\pi/4}(0) = 0$; $f(5\pi/4) = e^{5\pi/4}(-1/\sqrt{2} + 1/\sqrt{2}) = 0$.

$f'(x) = e^x(\sin x - \cos x) + e^x(\cos x + \sin x) = 2e^x \sin x = 0 \Rightarrow \sin x = 0$

$x = \pi \in (\pi/4, 5\pi/4) \Rightarrow c = \pi$. Rolle's theorem verified.

Problem 3: verify Rolle's theorem for $f(x) = e^{-x} \sin x$ on $[0, \pi]$.

$f(0) = 0 = f(\pi)$. $f'(x) = e^{-x}(\cos x - \sin x) = 0 \Rightarrow \tan x = 1 \Rightarrow c = \pi/4 \in (0, \pi)$ ✓.

Problem 4: verify Rolle's theorem for $f(x) = (x^2 + 3x)\, e^{-x/2}$ on $[-3, 0]$.

$f(-3) = 0 = f(0)$. $f'(x) = e^{-x/2}[\,2x + 3 - \tfrac{1}{2}(x^2 + 3x)\,] = \tfrac{1}{2}\,e^{-x/2}[\,4x + 6 - x^2 - 3x\,] = -\tfrac{1}{2}\,e^{-x/2}(x^2 - x - 6)$.

$f'(c) = 0 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0 \Rightarrow c = -2 \in (-3, 0)$ ($c = 3$ is outside).

Problem 5: verify Rolle's theorem for $f(x) = \log\left[\,(x^2 + ab)\,/\,(x(a + b))\,\right]$ on $[a, b]$ $(0 < a < b)$.

$f(a) = \log\left[\,(a^2 + ab)/(a(a+b))\,\right] = \log 1 = 0$; $f(b) = \log\left[\,(b^2 + ab)/(b(a+b))\,\right] = \log 1 = 0$.

$$f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x} = 0 \Rightarrow 2x^2 = x^2 + ab \Rightarrow x^2 = ab$$

$c = \sqrt{(ab)}$, the geometric mean, lies in (a, b) ✓.

Problem 6 (when it fails): is Rolle's theorem applicable to $f(x) = 1 - x^{2/3}$ on $[-1, 1]$?

$f(-1) = 0 = f(1)$ and f is continuous, but $f'(x) = -(2/3)x^{-1/3}$ does not exist at $x = 0 \in (-1, 1)$. Condition (ii) fails ⇒ Rolle's theorem is NOT applicable (indeed $f'(x) \neq 0$ anywhere).

Problem 7: show that between two roots of $e^x \sin x = 1$ there is at least one root of $e^x \cos x = -1$.

Let $g(x) = e^{-x} - \sin x$. $g(x) = 0 \Leftrightarrow e^{-x} = \sin x \Leftrightarrow e^x \sin x = 1$. If a, b are two roots, $g(a) = g(b) = 0$ and g is differentiable, so by Rolle there is $c \in (a, b)$ with $g'(c) = 0$: $-e^{-c} - \cos c = 0 \Rightarrow e^c \cos c = -1$. Hence the claim.

Problem 8: verify Rolle's theorem for $f(x) = \cos x$ on $[-\pi/2, \pi/2]$.

$f(-\pi/2) = 0 = f(\pi/2)$; $f'(x) = -\sin x = 0 \Rightarrow x = 0 \in (-\pi/2, \pi/2) \Rightarrow c = 0$ ✓.

Problem 9: prove that $x^3 - 3x + K = 0$ cannot have two distinct roots in $[0, 1]$ (K any constant).

Suppose it had two roots $a < b$ in $[0, 1]$. $f(x) = x^3 - 3x + K$ satisfies Rolle's conditions, so $f'(c) = 3(c^2 - 1) = 0$ for some $c \in (a, b) \subset (0, 1)$. But $c^2 - 1 < 0$ for $0 < c < 1$ – contradiction. So at most one root lies in

$[0, 1]$.

2. Lagrange's Mean Value Theorem (LMVT)

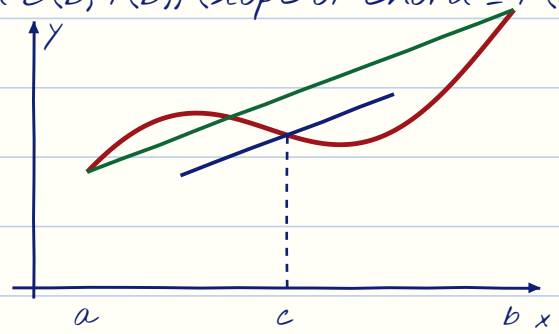
Statement: if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b), there exists $c \in (a, b)$ such that

$$f(b) - f(a) = (b - a)\, f'(c) \text{ i.e. } \frac{f(b) - f(a)}{b - a} = f'(c)$$

Alternative form: $f(a + h) = f(a) + h\, f'(a + \theta h)$, $0 < \theta < 1$ $(h = b - a)$.

Rolle's theorem is the special case $f(a) = f(b)$.

Geometrical meaning: there is a point on the curve where the tangent is PARALLEL to the chord joining the end points $A(a, f(a))$ and $B(b, f(b))$ (slope of chord $= f'(c)$).



Lagrange: chord AB

there is a point c where the tangent is PARALLEL to the chord AB

Problem 10: verify LMVT for $f(x) = x(x - 1)(x - 2)$ on $[0, 1/2]$.

$f(x) = x^3 - 3x^2 + 2x$; $f(0) = 0$; $f(1/2) = 1/2 \cdot (-1/2) \cdot (-3/2) = 3/8$. Slope of chord $= (3/8 - 0)/(1/2) = 3/4$.

$$f'(c) = 3c^2 - 6c + 2 = 3/4 \Rightarrow 12c^2 - 24c + 5 = 0$$

$$c = \frac{24 \pm \sqrt{(576 - 240)}}{24} = 1 \pm \frac{\sqrt{21}}{6}$$

$c = 1 - \sqrt{21}/6 \approx 0.2362 \in (0, 1/2)$ ✓ (the other root 1.76 is outside).

Problem 11: verify LMVT for $f(x) = \log x$ on $[1, e]$.

$f'(x) = 1/x$. Chord slope $= (\log e - \log 1)/(e - 1) = 1/(e - 1)$. $f'(c) = 1/c = 1/(e - 1)$

$\Rightarrow c = e - 1 \approx 1.718 \in (1, e)$ ✓.

Problem 12: find c of LMVT for $f(x) = x^3$ on $[1, 2]$.

$f(2) - f(1) = 8 - 1 = 7 = (2 - 1)\cdot 3c^2 \Rightarrow c^2 = 7/3 \Rightarrow c = \sqrt{(7/3)} \approx 1.5275 \in (1, 2)$ ✓.

Problem 13: find c for $f(x) = x^2 - 3x - 1$ on $[1117, 1317]$.

$f(b) - f(a) = (b^2 - a^2) - 3(b - a) = (b - a)(a + b - 3) \Rightarrow f'(c) = 2c - 3 = a + b - 3 \Rightarrow c = (a + b)/2 = 1217 \in (1117, 1317)$ ✓. (For ANY quadratic, c is the midpoint.)

Problem 14b: find c of LMVT for $f(x) = \sqrt{(x^2 - 4)}$ on $[2, 4]$.

$f(4) - f(2) = \sqrt{12} - 0 = 2\sqrt{3}$, chord slope $= \sqrt{3}$. $f'(x) = x/\sqrt{(x^2 - 4)} = \sqrt{3} \Rightarrow x^2 = 3x^2 - 12 \Rightarrow x^2 = 6 \Rightarrow c = \sqrt{6} \approx 2.449 \in (2, 4)$ ✓. (f is continuous at 2 and differentiable on the OPEN interval – enough for LMVT.)

Note: LMVT fails if the function is not differentiable inside, e.g. $f(x) = |x|$ on $[-1, 2]$: chord slope $= 1/3$ but $f' = \pm 1$ wherever it exists.

Applications – inequalities by LMVT:

Problem 15: prove $|\sin a - \sin b| \le |a - b|$.

Take $f(x) = \sin x$ on $[a, b]$: $\sin b - \sin a = (b - a)\cos c$. Since $|\cos c| \le 1$, $|\sin b - \sin a| \le |b - a|$.

Problem 16: prove $x/(1 + x) < \log(1 + x) < x$ for $x > 0$.

$f(t) = \log(1 + t)$ on $[0, x]$: $\log(1 + x) - 0 = x \cdot 1/(1 + c)$, $0 < c < x$. Since $1 < 1 + c < 1 + x$: $1/(1 + x) < 1/(1 + c) < 1$. Multiply by $x > 0$:

$$\frac{x}{1 + x} < \log(1 + x) < x$$

Problem 17: prove $(b - a)/(1 + b^2) < \tan^{-1}b - \tan^{-1}a < (b - a)/(1 + a^2)$, $0 < a < b$; hence $\pi/4 + 3/25 < \tan^{-1}(4/3) < \pi/4 + 1/6$.

$f(x) = \tan^{-1}x$: $\tan^{-1}b - \tan^{-1}a = (b - a)/(1 + c^2)$, $a < c < b$. As $a^2 < c^2 < b^2$, $1/(1 + b^2) < 1/(1 + c^2) < 1/(1 + a^2)$ – the result follows.

Take $a = 1$, $b = 4/3$: $b - a = 1/3$. Lower: $(1/3)/(1 + 16/9) = 3/25$; upper: $(1/3)/(1 + 1) = 1/6$. With $\tan^{-1}1 = \pi/4$:

$$\pi/4 + 3/25 < \tan^{-1}(4/3) < \pi/4 + 1/6$$

Problem 18: show that $e^x > 1 + x$ for $x > 0$ and that a function with $f'(x) = 0$ on an interval is constant.

(i) $f(t) = e^t$ on $[0, x]$: $e^x - 1 = x\,e^c > x$ since $e^c > 1$. (ii) For any $a < b$ in the interval, $f(b) - f(a) = (b - a)f'(c) = 0$, so $f(a) = f(b)$.

Problem 19: using LMVT prove $(b - a)/b < \log(b/a) < (b - a)/a$, $0 < a < b$.

$f(x) = \log x$ on $[a, b]$: $\log b - \log a = (b - a)/c$, $a < c < b$. Since $1/b < 1/c < 1/a$, multiply by $(b - a) > 0$: $(b - a)/b < \log(b/a) < (b - a)/a$. (For $a = 1$, $b = 1 + x$ this is the result of Problem 15.)

Problem 20: prove $\sin x < x$ for $x > 0$ and $\tan x > x$ for $0 < x < \pi/2$.

(i) $\sin x - \sin 0 = x \cos c \le x$ ($\cos c \le 1$; strict inequality since $\cos c < 1$ for $0 < c$). (ii) $\tan x - \tan 0 = x \sec^2 c > x$ since $\sec^2 c > 1$.

3. Cauchy's Mean Value Theorem

Statement: if $f(x)$ and $g(x)$ are (i) continuous on $[a, b]$, (ii) differentiable on (a, b), (iii) $g'(x) \neq 0$ for all x in (a, b), then there is $c \in (a, b)$ with

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Geometrically: for the curve $x = g(t)$, $y = f(t)$ there is a point where the tangent is parallel to the chord joining the end points. Taking $g(x) = x$ gives LMVT.

Problem 21: verify Cauchy's MVT for $f(x) = e^x$ and $g(x) = e^{-x}$ on $[a, b]$; show $c = (a + b)/2$.

$f'/g' = e^x/(-e^{-x}) = -e^{2c}$. LHS $= (e^b - e^a)/(e^{-b} - e^{-a}) = (e^b - e^a)\,e^{a+b}/(e^a - e^b) = -e^{a+b}$. So $e^{2c} = e^{a+b} \Rightarrow c = (a + b)/2 \in (a, b)$ ✓.

Problem 22: for $f(x) = \sqrt{x}$, $g(x) = 1/\sqrt{x}$ on $[a, b]$ show that c is the geometric mean $\sqrt{(ab)}$.

$f'/g' = [1/(2\sqrt{c})] / [-1/(2c^{3/2})] = -c$. LHS $= (\sqrt{b} - \sqrt{a})/(1/\sqrt{b} - 1/\sqrt{a}) = (\sqrt{b} - \sqrt{a})\sqrt{(ab)}/(\sqrt{a} - \sqrt{b}) = -\sqrt{(ab)}$. Hence $c = \sqrt{(ab)}$ ✓.

Problem 23: verify Cauchy's MVT for $f(x) = x^2$, $g(x) = x^3$ on $[1, 2]$.

LHS $= (4 - 1)/(8 - 1) = 3/7$. $f'/g' = 2c/(3c^2) = 2/(3c)$. $2/(3c) = 3/7 \Rightarrow c = 14/9 \approx 1.556 \in (1, 2)$ ✓.

Problem 24: verify Cauchy's MVT for $f(x) = \sin x$, $g(x) = \cos x$ on $[a, b]$.

LHS $= (\sin b - \sin a)/(\cos b - \cos a) = 2\cos((a+b)/2)\sin((b-a)/2) / [-2\sin((a+b)/2)\sin((b-a)/2)] = -\cot((a + b)/2)$. $f'/g' = \cos c/(-\sin c) = -\cot c. \Rightarrow c = (a + b)/2$ ✓.

Problem 25: using Cauchy's MVT show that $[f(b) - f(a)] / [g(b) - g(a)] = f'(c)/g'(c)$ reduces to LMVT for $g(x) = x$.

$g(b) - g(a) = b - a$ and $g'(c) = 1$, so $f(b) - f(a) = (b - a) f'(c)$.

Problem 26: for $f(x) = 1/x$, $g(x) = 1/x^2$ on $[a, b]$ find c.

$f'/g' = (-1/c^2)/(-2/c^3) = c/2$. LHS $= (1/b - 1/a)/(1/b^2 - 1/a^2) = ab/(a + b)$. So $c = 2ab/(a + b)$, the HARMONIC mean of a and b ✓.

Problem 27: for $f(x) = x$, $g(x) = x^2$ on $[a, b]$ find c.

LHS $= (b - a)/(b^2 - a^2) = 1/(a + b)$; $f'/g' = 1/(2c) \Rightarrow c = (a + b)/2$, the arithmetic mean ✓.

4. Taylor's and Maclaurin's Theorems (without proof)

Taylor's theorem: if f and its derivatives up to order $n-1$ are continuous on $[a, x]$ and $f^{(n)}$ exists on (a, x), then

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

$$R_n = \frac{(x-a)^n}{n!}f^{(n)}(c),\ a < c < x \quad \text{(Lagrange's form of remainder)}$$

Maclaurin's theorem: put $a = 0$:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + R_n$$

$$R_n = \frac{x^n}{n!}f^{(n)}(\theta x),\ 0 < \theta < 1$$

Standard Maclaurin series:

Function	Series	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + \dots$	all x
$\sin x$	$x - x^3/3! + x^5/5! - \dots$	all x
$\cos x$	$1 - x^2/2! + x^4/4! - \dots$	all x
$\log(1+x)$	$x - x^2/2 + x^3/3 - x^4/4 + \dots$	$-1 < x \le 1$
$(1+x)^m$	$1 + mx + m(m-1)x^2/2! + \dots$	$\lvert x\rvert < 1$
$\tan^{-1}x$	$x - x^3/3 + x^5/5 - \dots$	$\lvert x\rvert \le 1$
$\sinh x$, $\cosh x$	$x + x^3/3! + \dots$, $1 + x^2/2! + \dots$	all x

Problem 28: expand e^x by Maclaurin's theorem with Lagrange's remainder.

$f^{(K)}(x) = e^x \Rightarrow f^{(K)}(0) = 1$ for all K.

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!} + \frac{x^n}{n!} e^{\theta x}$$

Since $x^n/n! \to 0$ as $n \to \infty$, $R_n \to 0$ and the infinite series converges to e^x for every x.

Problem 29: expand $f(x) = x^3 - 2x^2 + 3x + 5$ in powers of $(x - 2)$.

$f(2) = 11$; $f' = 3x^2 - 4x + 3 \to f'(2) = 7$; $f'' = 6x - 4 \to f''(2) = 8$; $f''' = 6$.

$$f(x) = 11 + 7(x - 2) + \frac{8}{2!}(x - 2)^2 + \frac{6}{3!}(x - 2)^3$$

$$\boxed{= 11 + 7(x - 2) + 4(x - 2)^2 + (x - 2)^3}$$

Problem 30: expand $\log x$ in powers of $(x - 1)$ up to four terms.

$f = \log x$, $f' = 1/x$, $f'' = -1/x^2$, $f''' = 2/x^3$, $f'''' = -6/x^4$; at $x = 1$: $0, 1, -1, 2, -6$.

$$\boxed{\log x = (x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots}$$

Problem 31: expand $\sin x$ in powers of $(x - \pi/2)$.

Derivatives at $\pi/2$: $\sin = 1$, $\cos = 0$, $-\sin = -1$, $-\cos = 0$, $\sin = 1$ …

$$\sin x = 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \dots$$

Problem 32: find $\sin 31°$ (approximately) using Taylor's theorem about $\pi/6$.

$a = 30° = \pi/6$, $h = 1° = \pi/180 = 0.017453$. $f(a + h) = f(a) + h\,f'(a) + (h^2/2)\,f''(a) + (h^3/6)\,f'''(a)$ with $\sin \to \cos \to -\sin \to -\cos$:

$$\sin 31° \approx \tfrac{1}{2} + (\sqrt{3}/2)(0.017453) - \tfrac{1}{2} \cdot \frac{(0.017453)^2}{2} - (\sqrt{3}/2)\frac{(0.017453)^3}{6}$$

$$\boxed{= 0.5 + 0.015115 - 0.000076 - 0.0000008 \approx 0.51504}$$

Problem 33: find $\sqrt{4.1}$ approximately using Taylor's expansion of $\sqrt{x}$ about $x = 4$.

$f=\sqrt{x}$: $f(4)=2$, $f'=1/(2\sqrt{x})\to 1/4$, $f''=-1/(4x^{3/2})\to -1/32$, $f'''=3/(8x^{5/2})\to 3/256$.

$$\sqrt{4+h}=2+\frac{h}{4}-\frac{h^2}{64}+\frac{h^3}{512}-\dots$$

$h=0.1$: $2+0.025-0.000156+0.000002=2.02485$ (true value $2.024845\ldots$).

Problem 34: expand $e^x\sin x$ up to the term in x^5 (Maclaurin).

Multiply $e^x=1+x+x^2/2+x^3/6+x^4/24+x^5/120$ and $\sin x=x-x^3/6+x^5/120$:

$$\boxed{e^x\sin x=x+x^2+\frac{x^3}{3}+0\cdot x^4-\frac{x^5}{30}+\dots}$$

Problem 35: expand $\tan x$ up to x^5; and $e^{\sin x}$ up to x^4.

$$\tan x=x+\frac{x^3}{3}+\frac{2x^5}{15}+\dots$$

$$e^{\sin x}=1+x+\frac{x^2}{2}-\frac{x^4}{8}+\dots$$

Problem 36 (limits by series): evaluate $\lim (x-\sin x)/x^3$ and $\lim (\tan x-x)/x^3$ as $x\to 0$.

$x-\sin x=x^3/6-x^5/120+\dots\Rightarrow (x-\sin x)/x^3\to 1/6$. $\tan x-x=x^3/3+2x^5/15+\dots\Rightarrow(\tan x-x)/x^3\to 1/3$.

Problem 37 (applications with remainder): show that $\cos x\ge 1-x^2/2$ and $\sin x>x-x^3/6$ for $x>0$.

Maclaurin with 3 terms: $\cos x=1-x^2/2+(x^4/4!)\cos\theta x$ and $(x^4/24)\cos\theta x$ can be negative, but the 2-term form $\cos x=1-(x^2/2)\cos\theta x\ge 1-x^2/2$ for $x^2\ge 0$. Similarly $\sin x=x-x^3/6+(x^5/120)\cos\theta x$, and for $0<x<\pi/2$ the remainder is positive, so $\sin x>x-x^3/6$.

Problem 38: expand $\cos x$ by Maclaurin with remainder; and $(1+x)^{1/2}$ up to x^3.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{x^{2n}}{(2n)!}\cos(\theta x + n\pi)$$

$$\sqrt{(1+x)} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots$$

Problem 39: expand $\sin x$ in powers of $(x - \pi/4)$ up to the cubic term.

At $\pi/4$: $\sin = \cos = 1/\sqrt{2}$; $f' = \cos$, $f'' = -\sin$, $f''' = -\cos$.

$$\sin x = \frac{1}{\sqrt{2}}\left[1 + (x - \pi/4) - \frac{(x-\pi/4)^2}{2!} - \frac{(x-\pi/4)^3}{3!} + \dots\right]$$

Problem 40: expand e^x in powers of $(x - 1)$ and $\sec x$ up to x^4.

$$e^x = e\left[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \dots\right]$$

$$\sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Problem 41 (more limits): evaluate $\lim (e^x - e^{-x} - 2x)/(x - \sin x)$ and $\lim (\cos x - 1 + x^2/2)/x^4$.

$e^x - e^{-x} - 2x = x^3/3 + \dots$, $x - \sin x = x^3/6 - \dots \Rightarrow$ ratio $\to (1/3)/(1/6) = 2$. $\cos x - 1 + x^2/2 = x^4/24 - \dots \Rightarrow$ limit $= 1/24$.

Problem 42 (error estimate): find the number of terms of the series of e^x needed to compute e to within 0.001.

$R_n = (1/n!)\,e^\theta < 3/n!$ for $x = 1$. Need $3/n! < 0.001 \Rightarrow n! > 3000 \Rightarrow n = 7$ ($7! = 5040$). So the first 7 terms (up to $1/6!$) give $e \approx 2.7181$ with error < 0.001.

Summary of the three mean value theorems:

Theorem	Conditions	Conclusion	Geometry
Rolle	f cont. on $[a,b]$, diff. on (a,b), $f(a) = f(b)$	$f'(c) = 0$	horizontal tangent
Lagrange	f cont. on $[a,b]$, diff. on (a,b)	$f(b) - f(a) = (b-a)\,f'(c)$	tangent ∥ chord
Cauchy	f, g cont. on $[a,b]$, diff. on (a,b), $g' \neq 0$	$[f(b)-f(a)]/[g(b)-g(a)] = f'(c)/g'(c)$	parametric curve
Taylor	f has derivatives up to order n	$f(x) = \Sigma\,(x-a)^k\,f^{(k)}(a)/k! + R_n$	polynomial approximation

Important Questions – Unit III

1. State and verify Rolle's theorem for given functions (problems 1–5); when does it fail?
2. State Lagrange's MVT, give its geometrical meaning, verify it and find c.
3. Prove inequalities using LMVT (problems 14–19).
4. State Cauchy's MVT and verify for pairs of functions.
5. State Taylor's and Maclaurin's theorems; expand e^x, $\sin x$, $\cos x$, $\log(1 + x)$.
6. Expand a given function in powers of $(x - a)$; approximate values ($\sin 31°$, $\sqrt{4.1}$).
7. Use Maclaurin series to evaluate limits and to expand products such as $e^x \sin x$, $\tan x$.